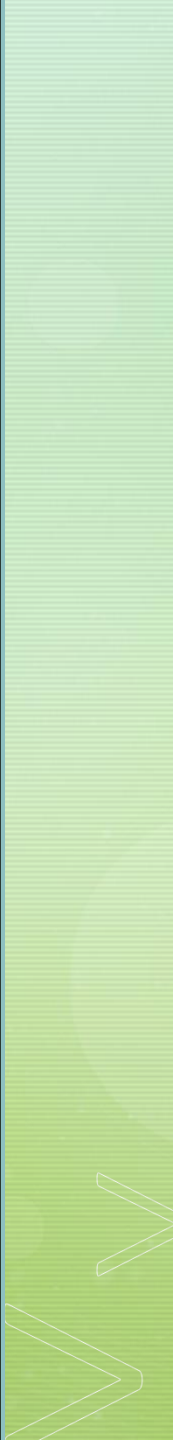




Ethics in Digital Forensics

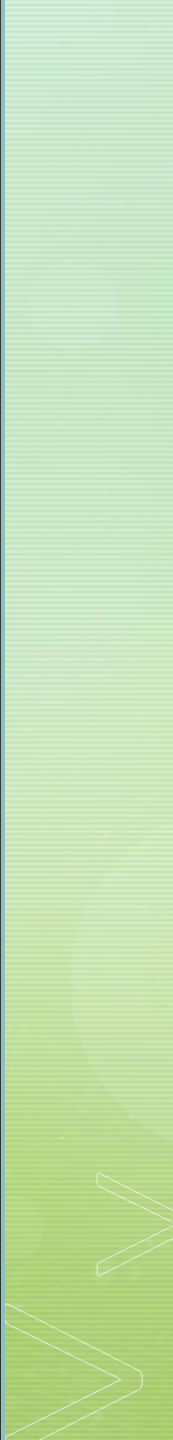


What is Digital Forensics?

- Definition: Investigating digital devices to collect evidence
 - Applications: Cybercrime, corporate security, legal cases
- 

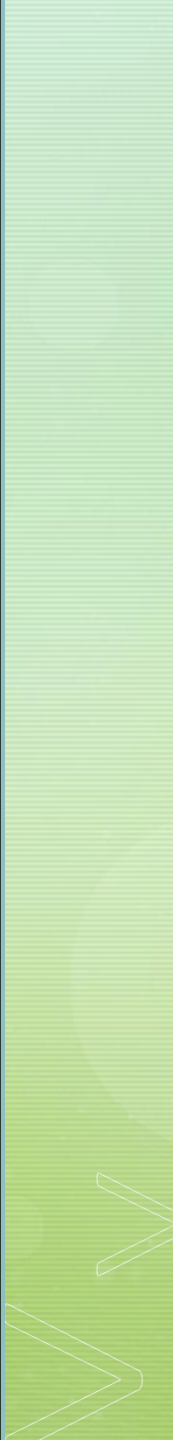


Why Ethics Matters

- Sensitive personal data involved
 - Risk of privacy violations
 - Maintaining legal admissibility and fairness
- 

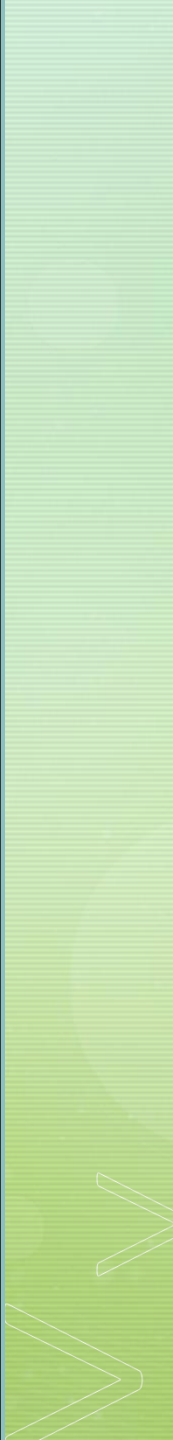


Core Ethical Principles

- Integrity of evidence
 - Confidentiality and privacy
 - Impartiality and objectivity
 - Accountability
- 

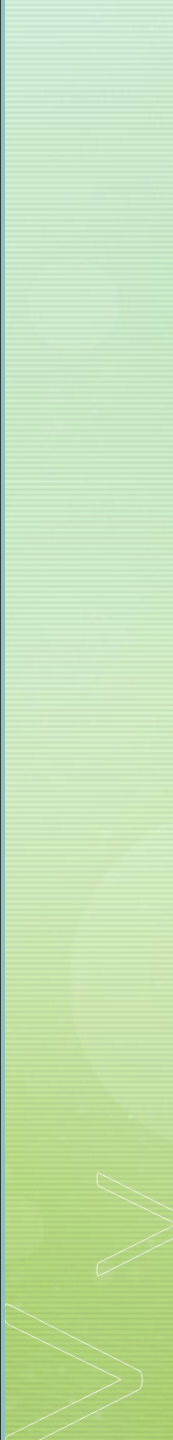


Legal Frameworks

- Computer Fraud and Abuse Act (CFAA)
 - GDPR for data privacy
 - Chain of custody requirements
- 



Common Ethical Dilemmas

- Accessing personal communications without consent
 - Handling privileged or confidential data
 - Using hacking tools during investigations
- 

Case Study 1: BTK Killer (2005)

- Metadata from Word document on floppy disk led to arrest
- Question: Was accessing deleted files ethical and legal?

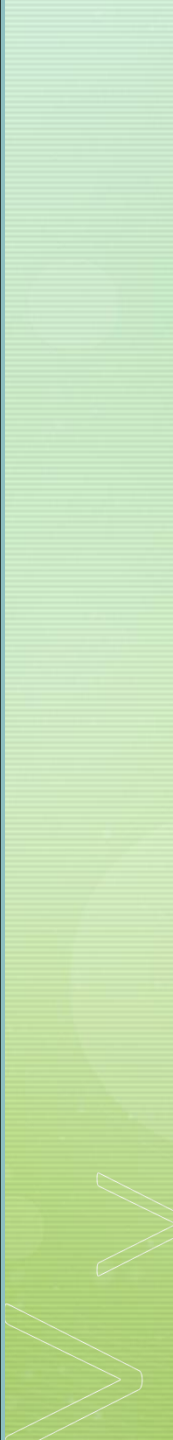


Case Study 2: Enron Email Investigation

- Massive email recovery for fraud evidence
- Issues: Employee privacy vs. corporate accountability

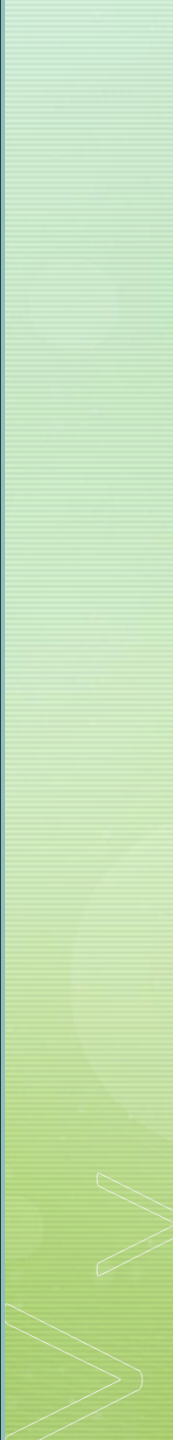


Case Study 3: iPhone San Bernardino (2016)

- FBI vs Apple encryption battle
 - Debate: National security vs user privacy
- 



Professional Codes of Conduct

- ISFCE Code of Ethics
 - ACM and IEEE ethical guidelines
 - Principle: 'Do no harm' and respect privacy
- 



Privacy vs Justice Debate

- Should investigators access all data for solving crimes?
- Where do we draw the line for personal privacy?

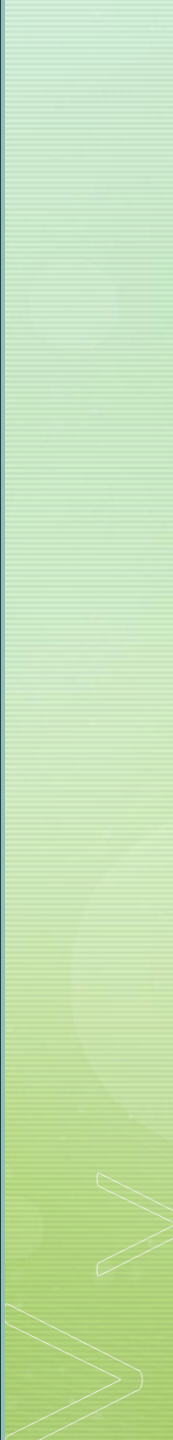


Activity: Class Debate

Divide into two groups:

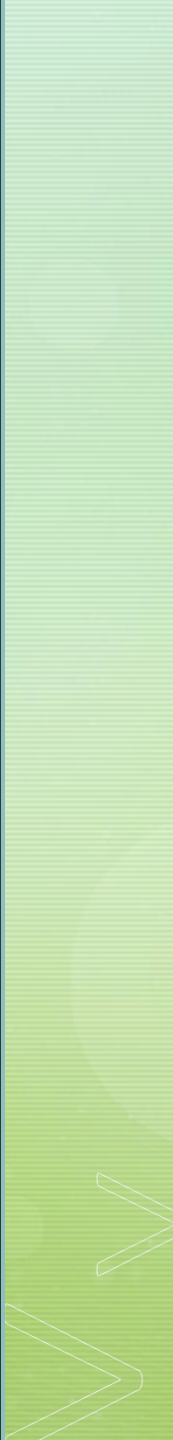
- Group A: Prioritize national security over privacy.
- Group B: Prioritize user privacy over government access.

“Prepare 3 key arguments on each”





Challenges in Digital Forensics

- Encryption and anti-forensic tools
 - Cloud data jurisdiction issues
 - Big Data and AI complicating analysis
- 



Ethical Use of Forensic Tools

- Avoid planting evidence
- Document every step
- Use tools lawfully and transparently

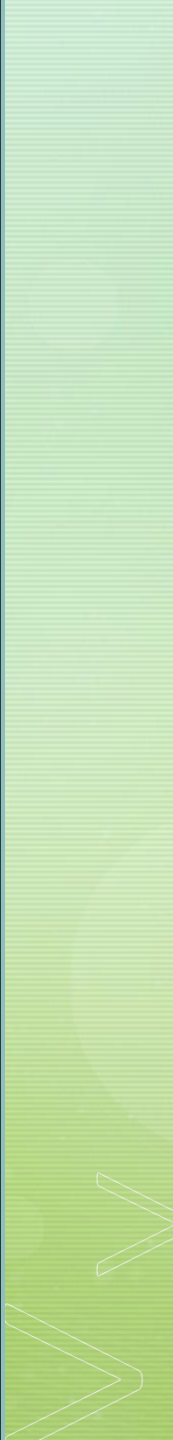


Handling Child Exploitation Cases

- Extreme psychological and ethical burden
- Need for controlled environments
- Mandatory reporting obligations



Corporate Investigations

- Employee monitoring ethics
 - Balancing IP protection with privacy
 - Avoiding misuse of investigation powers
- 



Scenario Analysis

Scenario: You find unrelated illegal content during a corporate investigation.

What steps should you take?

Discuss in pairs and justify your decision.

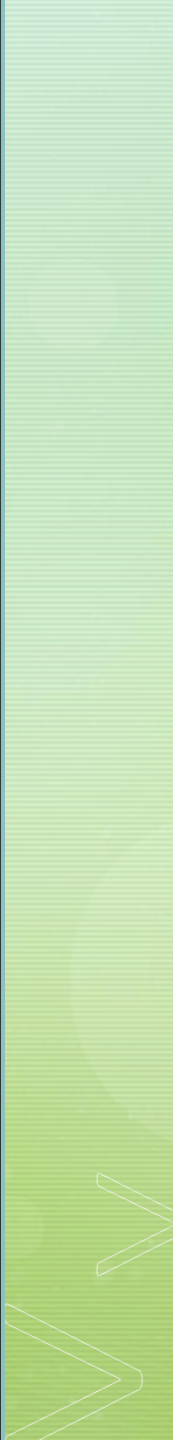


Emerging Trends & Ethical Risks

- Cloud and cross-border challenges
- AI-assisted forensics and bias
- Deepfake evidence manipulation



Best Practices

- Always maintain chain of custody
 - Seek warrants where required
 - Follow professional codes and training
- 



Quick Quiz

1. What is the chain of custody?
2. Name two professional ethics codes in forensics.
3. Give one ethical dilemma in mobile device forensics.



Summary & Takeaways

- Ethics ensures trust, legality, and justice
 - Respect privacy but uphold law
 - Critical thinking required in every case
- 